

júliabiró

infrastructure & reliability engineer

about

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languages

fluent English
native Hungarian
B1 German

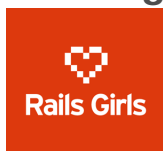
code

go python
javascript bash

dev

Terraform, Kubernetes,
AWS, Fastly VCL,
nginx, everything
configuration

teaching



django girls



interests

Reliability, scalability, security, monitoring, incident management

education

- 2007 – 2013 **BSc degree in IT Engineering/CS** specialization in software development
Budapest University of Technology and Economics
- 2002 – 2010 **MSc degree in Mathematics** specialization in operation theory
Eötvös Loránd University of Sciences

experience

- 2017-2025 **Contentful** reliability team
senior reliability engineer
- 2014 – 2017 **Prezi** infrastructure team
infrastructure and tooling engineer
- 2011 – 2014 **Ericsson Hungary** edge router software testing
software developer
- 2008 – 2011 **MTA SZTAKI** Data-mining and Web search team
software developer

skills

Infrastructure management
AWS, Fastly, Kubernetes, Terraform

Monitoring
Prometheus, Grafana, Splunk, OpenTelemetry

Reliability
Networking, procedures

Caching
Fastly, Varnish, Akamai, CloudFront

scope

team
roadmap, project lead, implementation

group
collaboration on initiatives, establishing processes

across group limits
consulting on implementation

all engineering
establishing processes

across department limits
knowledge sharing, internal enablement

hobbies

European cinema; museums; reading quality journalism

projects

driver

Migrations

I assisted in the execution and planning of several migrations projects, with regards to service availability, data integrity, cache behavior etc. Examples are spread out around: I've seen/done/planned migrations between secret storage solutions, monitoring and alerting providers, application platforms, cache invalidation strategies, async message bus providers, WAF providers, edge cache providers. Even the move of a small stateless service between clouds.

Monitoring

Maintainer and later driver of monitoring. Did everything from editing Grafana dashboards to mapping out the current state and defining a future strategy for monitoring and observability. Multiple rounds of moving/restructuring pipelines for alerting.

Reliability

Contributed to implementing multi-region service of stateful API. Advising on migration plans for an async communication flow between two message bus systems. Defining and initiating process for managing reliability-related risks from a business perspective. Reliability/scalability evaluation before events where changing or more critical traffic is expected. Multiple projects for reducing blast radiuses by routing and sectioning traffic.

Incident management

Introduced, defined and maintained incident management processes to Contentful (documentation, knowledge sharing, leading by example, helping postmortems, internal blogging etc). I have been on call for 10 years, I have participated in countless incidents, even customer incidents. Written customer-facing Root Cause Analyses.

expert

Caching

Several years of experience debugging edge layer issues, including connectivity and stale cache issues Internal knowledge building about caching and cache invalidation, enabling Customer Support team to handle more technical tickets Planning workflow for changing cache invalidation strategy (to avoid stale cache) Customer-facing product maintenance for caching-related products.

Operations

Operating an application platform layer of dozens of Kubernetes clusters Managing a fleet of several hundred instances of RabbitMQ async message queues (but you can't pay me enough to do this again :)) Evaluating multi-region database backup solutions. Maintaining and developing internal and external routing.

Security

Contributed to multiple projects where we changed how we store and access secrets. One of the maintainers of <https://github.com/contentful-labs/kube-secret-syncer>, a tool for syncing AWS Secrets Manager contents to Kubernetes Secrets. I was a "Security Champion", an in-team representative of security-awareness. Pointed out multiple times when internal information was shared publicly by accident

Architecture

Mapping out the state of internal routing, and possible future directions Mapping out the state of the organically diverse application platform, suggesting future directions for keeping it the simplest possible. Advising on several projects where data is distributed in several layers of the architecture, or where failure domains or blast radiuses would change. Observed and adapted to a 100x change in infrastructure scale

experience

Compliance

Defining, maintaining and improving the Disaster Recovery Procedure, both the technical and the customer-facing aspects. Translating user access audit requirements for infrastructure tools and services for SOC2 compliance

Product-oriented

On multiple occasions I contributed to the structured product definition and maintenance of features that started as a “one-off” but then got rolled out more broadly. This included internal and external documentation and knowledge sharing, pointing out upcoming product questions, interactions with other features, measuring business impact etc.